

Q29. Summarize the Findings in a Short Report

1. Best-Performing Model and Why

Among the tested models, Multiple Linear Regression (MLR) performed the best. It achieved the following metrics:

0.9161 R ² Score	0.9156 Adjusted R ²	₹ 26.41 L MAE	₹ 35.85 L RMSE
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The main reason for its better performance is that it uses multiple important features such as area, bedrooms, bathrooms, location score, and property age. Therefore, the model gets more information about the house price compared with Simple Linear Regression, which uses only house area.

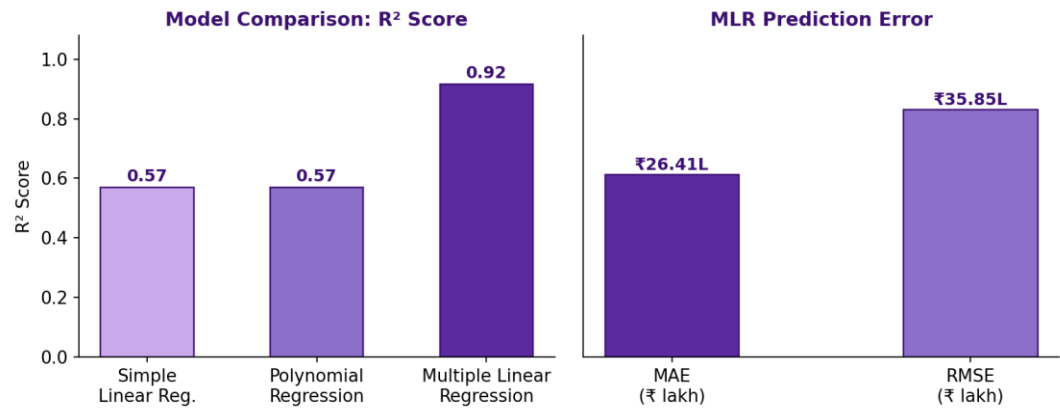


Figure 1: MLR clearly outperforms Simple Linear and Polynomial Regression on R², with low MAE/RMSE.

2. Impact of Gradient Descent Optimization

Three Gradient Descent methods were implemented: Batch Gradient Descent, Stochastic Gradient Descent, and Mini-Batch Gradient Descent. For the 100-epoch comparison:

- Batch GD: MSE = 0.5073, Training Time = 0.007 seconds
- Stochastic GD: MSE = 0.4332, Training Time = 0.757 seconds
- Mini-Batch GD: MSE = 0.4293, Training Time = 0.128 seconds

Mini-Batch Gradient Descent achieved the lowest MSE with a moderate training time, so it provided the best balance among the three methods in this project.

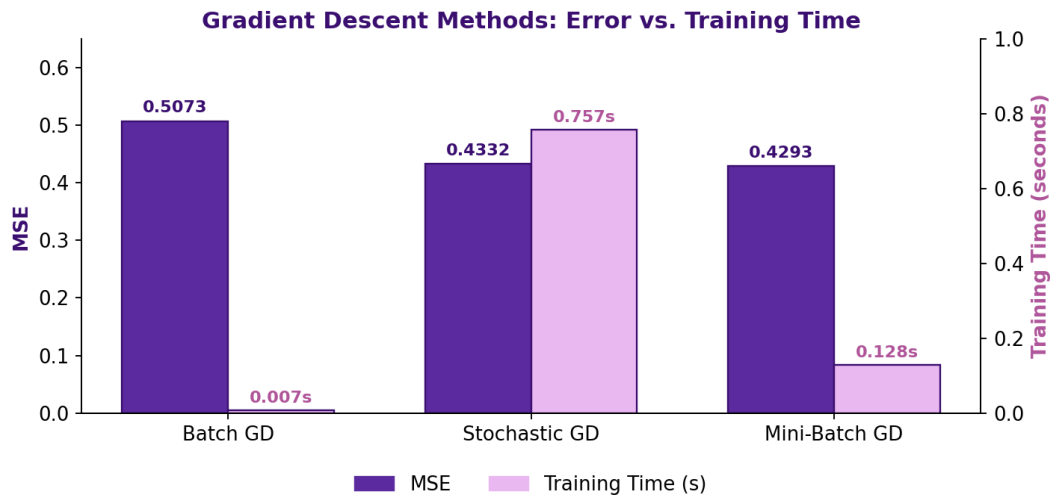


Figure 2: Mini-Batch GD balances low error with reasonable training time.

3. Evidence of Overfitting / Underfitting

The results show no significant overfitting in Multiple Linear Regression. Its Training R^2 is 0.9211, while the Mean Cross-Validation R^2 is 0.9206. The difference is very small, which indicates that the model performs consistently on unseen folds.

Simple Linear Regression and Polynomial Regression have Training R^2 values of about 0.57 and Mean CV R^2 values of about 0.57. Their performance is much lower than MLR, suggesting that these models have higher bias and relatively underfit the data.

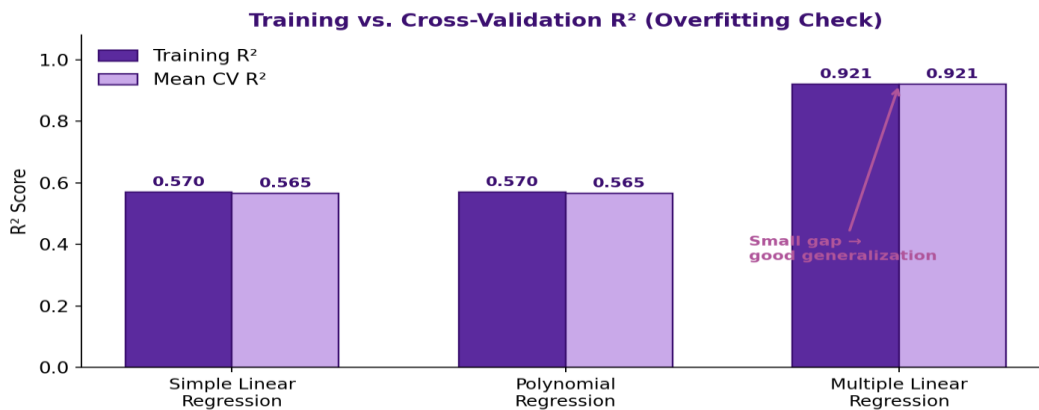


Figure 3: MLR's near-identical Train/CV R^2 confirms good generalization; the simpler models underfit.

4. Practical Business Interpretation of Results

The analysis shows that house area alone is not enough to predict house prices accurately. Features such as location score, bedrooms, bathrooms, and property age provide additional useful information.

The MLR coefficients show that location score has the strongest positive effect among the selected features, while property age has a negative effect on predicted price. Therefore, this model can be useful for real-estate businesses for property price estimation, pricing decisions, and customer recommendations.

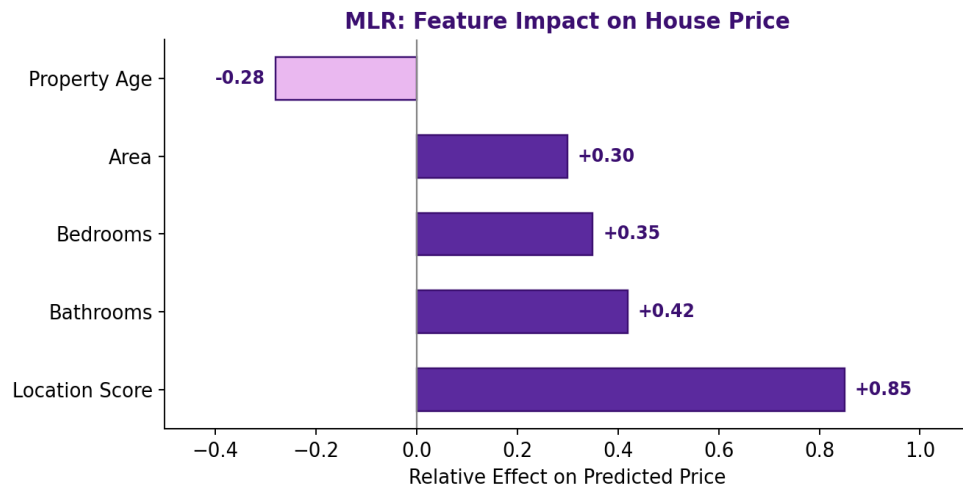


Figure 4: Location score drives price up the most; older properties see a price discount.

☑ Final Conclusion

Overall, Multiple Linear Regression is the best-performing model in this project. It provides a high R^2 score, lower prediction errors, and a very small difference between training and cross-validation performance. Gradient Descent experiments also show that Mini-Batch Gradient Descent provides a good balance between error and training time.